CSC-335

HCI & Computer Graphics

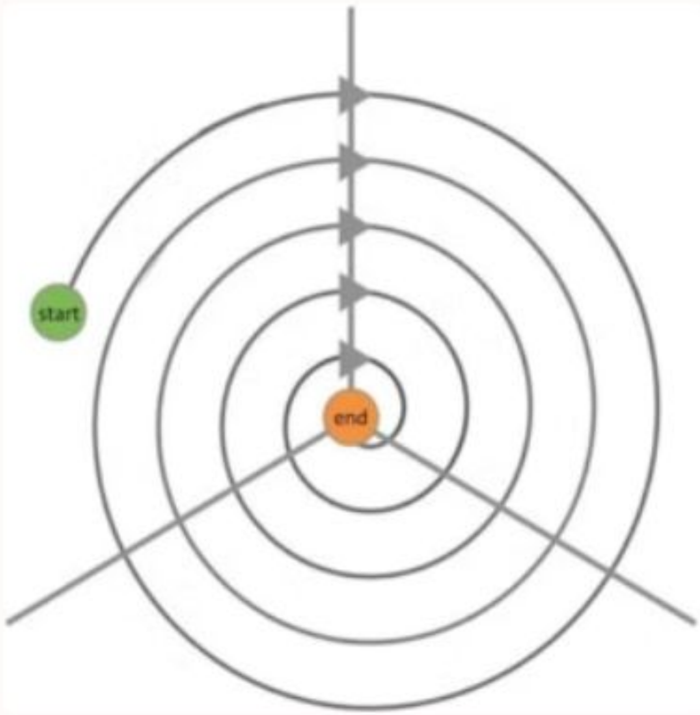
Lec 4: UX Components and User Testing

Methods of UX Research

Three Foundational Methods of UX Research

ASK	OBSERVE	INSPECT
Understand users' perspectives by asking them directly about their thoughts, needs, and behaviors.	Gain insights by watching users interact with a system or perform tasks in their natural environment.	Evaluate designs and prototypes against established guidelines and best practices to identify potential usability issues.

Comprehensive UX Research Framework



Assess

Interviews, Surveys, Comparative Analysis

Design

Lo-Fi Testing, Hi-Fi Testing

Build

Heuristic Evaluation, Analytics, Observations

A multi-faceted approach combines various techniques across the product development lifecycle to ensure a holistic understanding of user needs and system performance.

The 3 Methods

ASK

OBSERVE

INSPECT

Ask: Gathering User Perspectives

Interviews

Structured conversations with stakeholders to deeply understand their experiences, motivations, and challenges.

Surveys

Distributed questionnaires to a large audience to collect quantitative and qualitative data on attitudes, behaviors, and demographics.

Focus Groups

Facilitated discussions with small groups of users to explore specific topics and gather collective opinions.

Diary Studies

Participants record their experiences, thoughts, and activities over an extended period to capture real-time usage and context.

Experience Sampling

Collecting data on users' experiences at random intervals in their natural environment, often via mobile devices.

Observe: Understanding User Behavior

Ethnographic Observations

Watching users in their natural environment to understand how they interact with products and perform tasks without intervention.

User Testing

Observing users as they complete specific, scripted tasks within a system to identify usability issues and assess task completion rates.

Usage Analytics

Analyzing large datasets of user interactions (e.g., clicks, navigation paths) to uncover patterns and trends in system usage.

Video Analysis

Reviewing recorded user sessions to meticulously analyze interactions, gestures, and expressions, providing rich qualitative data.

Social Media Mining

Extracting and analyzing publicly available data from social media to understand user sentiment, feedback, and trends related to a product.

Inspect: Evaluating Designs

Guideline-Based

Systematic evaluation of a design against established usability heuristics, accessibility standards, or industry best practices.

Walkthroughs

Simulating user interaction paths step-by-step to uncover potential friction points or breakdowns in the user journey.

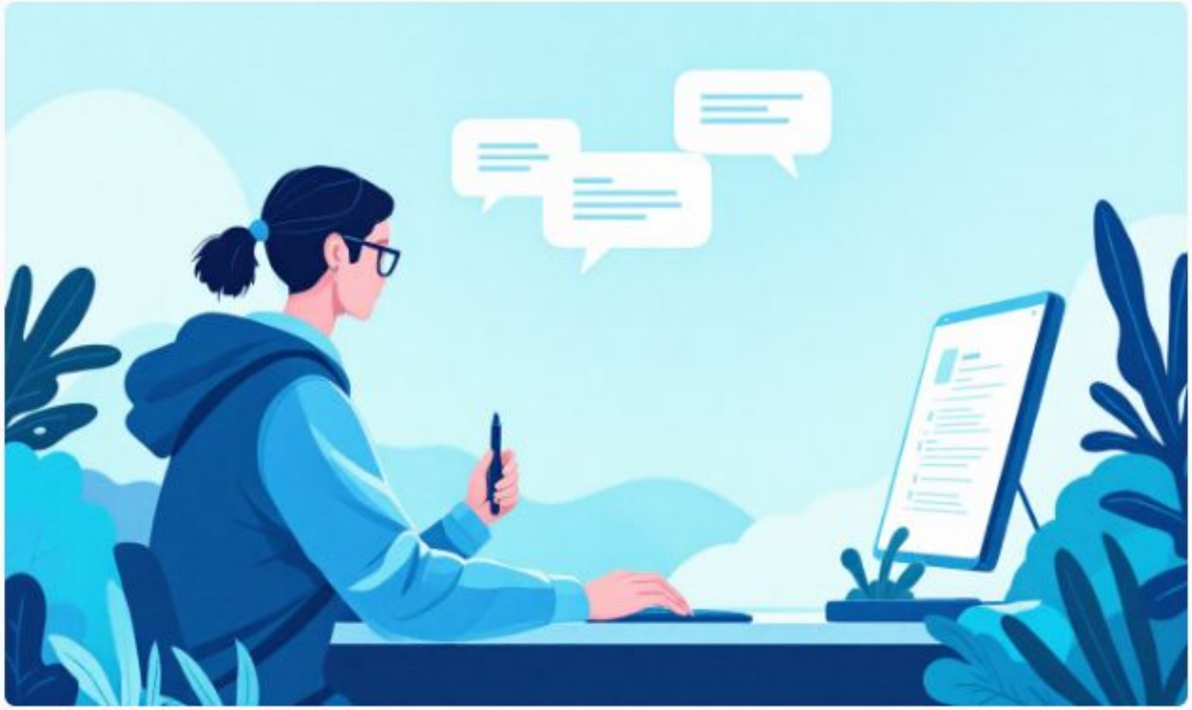
Comparative Analysis

Assessing a product's design by comparing it against competitors or similar solutions to pinpoint strengths, weaknesses, and opportunities.

Hybrid Approaches: Watch and Ask

Often, combining observation with direct questioning yields richer insights, capturing both explicit and implicit user behaviors and motivations.

- **User Testing + Interviews**Observing users perform tasks, then asking follow-up questions to understand their decision-making and rationale.
- **Contextual Interviews**Interviewing users in their natural environment while they are engaging in relevant activities, capturing insights in real-time context.
- **Artifact-based Methods**Analyzing existing user-generated artifacts (e.g., notes, sketches, logs) to understand workflows and challenges, often combined with interviews about these artifacts.



When to Apply Which Method?

Ask when...

- Observation is infeasible due to infrequent, long-duration, or private activities.
- Understanding user values, motivations, and emotional responses is key.
- Large numbers and high certainty are needed (e.g., through surveys).

Observe when...

- Understanding actual behavior, not just reported behavior, is crucial.
- Identifying usability issues in context is important.
- Discovering unforeseen usage patterns is desired.

Inspect when...

- There is a need for a quick and cost-effective expert evaluation.
- Benchmarking against established standards or competitors is required.
- Identifying probable flaws before extensive user testing is beneficial.


WHAT MAKES A GOOD
USER EXPERIENCE?
01 STEP
PANTONE
&
TOOL

Components of UX

Good and Bad UX

Usef
ul

Stressf
ul

Inefficient

Confusi
ng

Easy to
Learn

Fu
n

Ugly

Distracting

Helpf
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Connect
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Tedious

Satisfyi
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Frustrati
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Condescend

Inconsider
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Accessib
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Attracti
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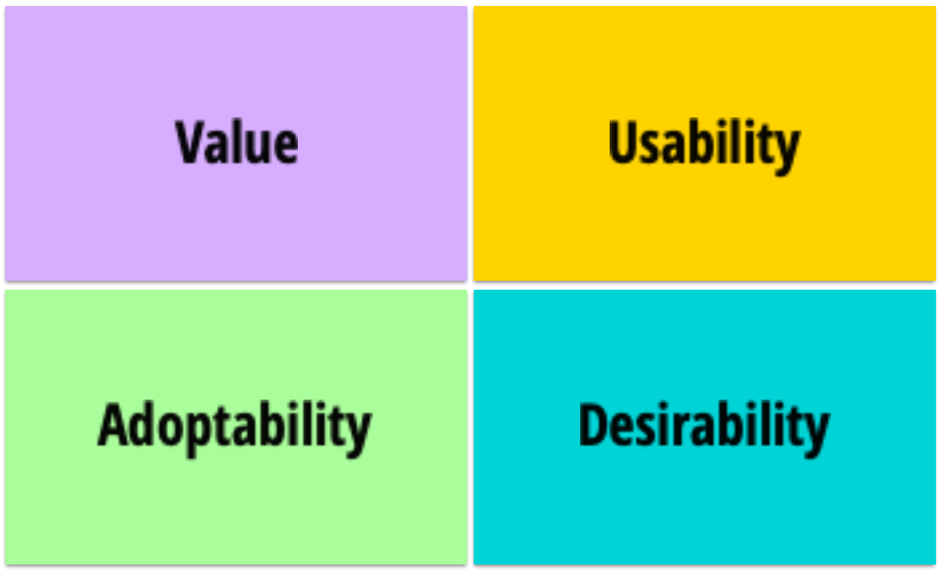
Delightful

Components of UX



Components of UX

VisiCalc 1979

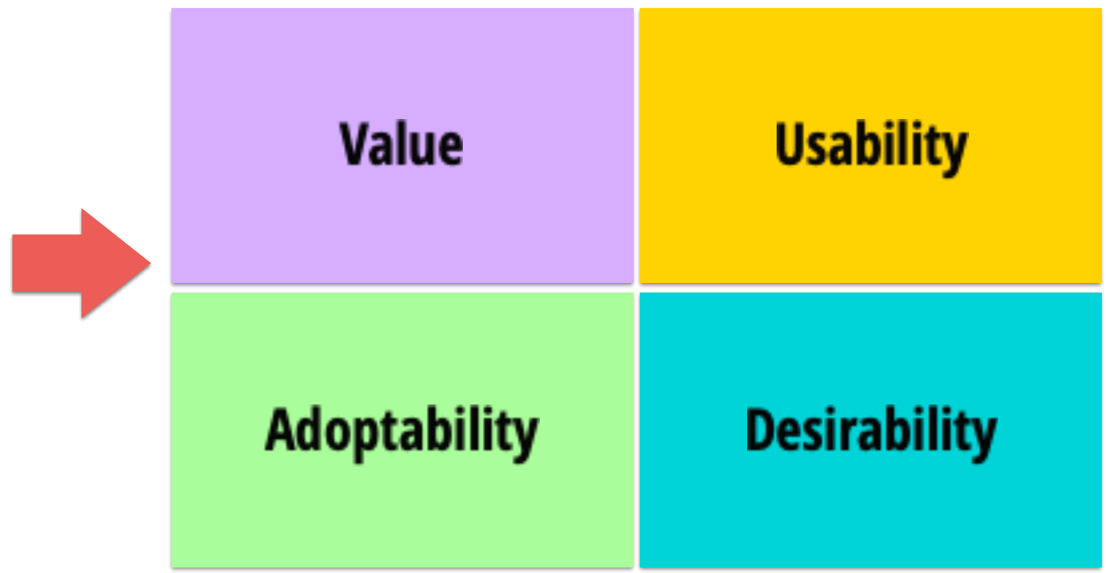
MOTOROLA ROKR

iPod 2001

Adoptability - Duolingo Onboarding Process

Basic Methods of UX

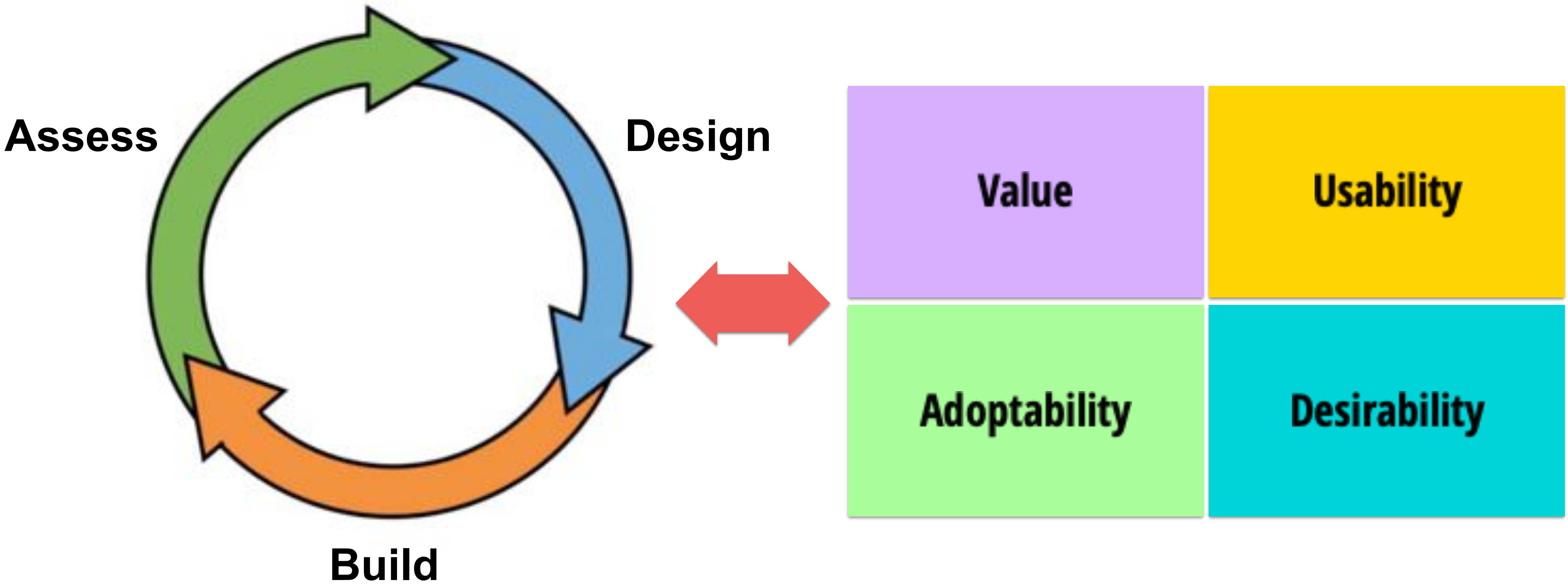
- Understanding Users
- Design & Prototyping
- Evaluating Designs



Assessing UX: Questions

	Understanding Users	Evaluating Designs
Value	What do users need?	Does this design fulfill the need?
Usability	How do they do it now?	Can they get it done with this?
Desirability	What do they desire?	Is the design appealing?
Adoptability	Where do users look for things?	Can users find and access this?

The Process is your Guide!



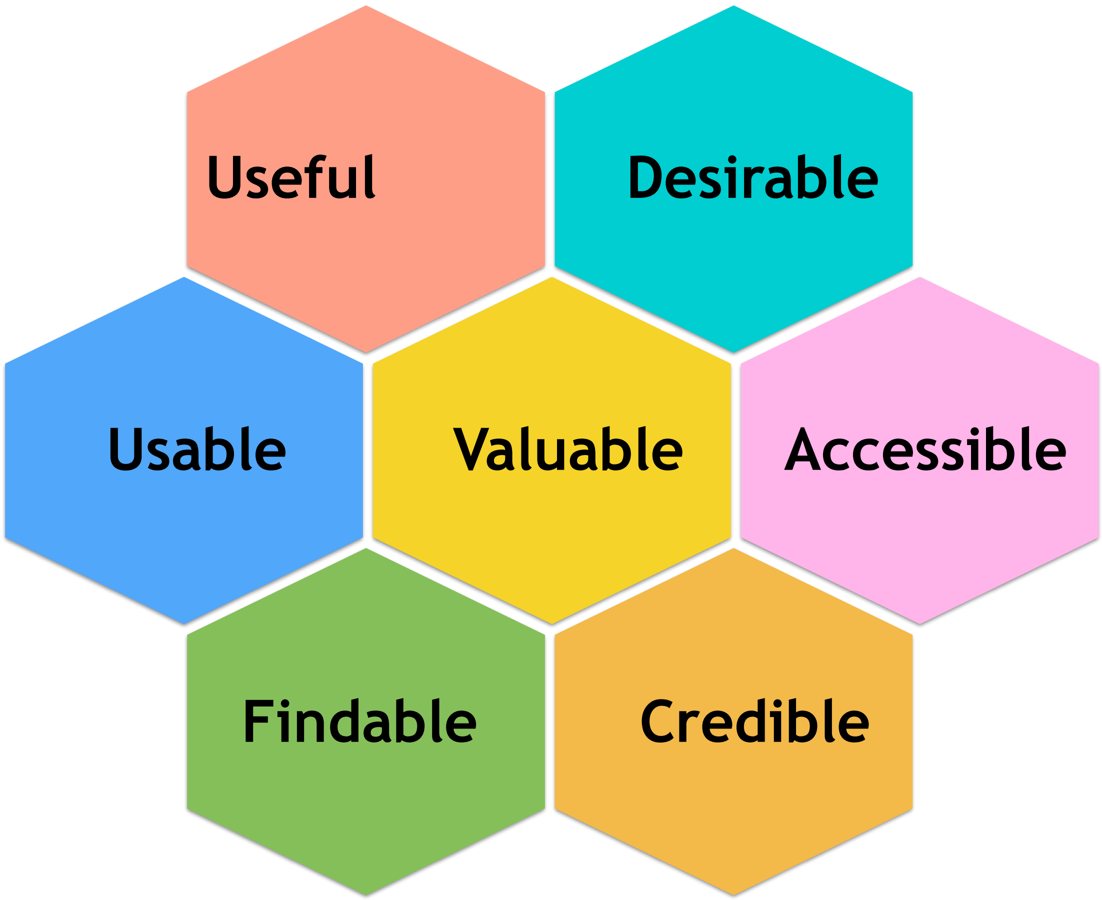
The 7 Factors that influence User Experience: https://www.interaction-design.org/literature/article/the-7-factors-that-influence-user-experience

The UX Honeycomb

Peter Morville

The UX Honey Comb: http://semanticstudios.com/user_experience_design/

The UX Honeycomb



User Testing

Micro-Usability Test

Jump Right In...

- User testing is one of the core methods in UX research, if not the core method in UX research.

- Do some User Testing!

What is User Testing?

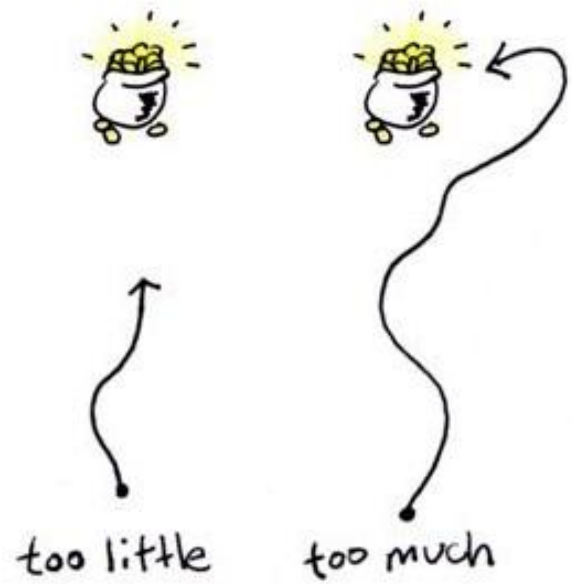
- **Watch representative users, try to accomplish important tasks using that product.**
  - *Representative Users: the types of people that you expect to use a system*
- **AKA Usability Testing**
  - *Theres more to UX than just Usability Testing*
  -

Why User Testing?

- So, why do we do user testing?
- What do you get out of it?
- You learn a *lot* from watching people use a system.
  - *What works and what doesn’t?*
  - *Why things work and don’t?*
  - *User needs that you missed*

Why User Testing?

- **Why not just use your own experience?**
  - *You often know too much*
  - *You might know too little*



Basic Idea of User Testing

- **Plan the Usability Test**
- **Find Potential Users**
- **Ask them to do some stuff**
  - *Tasks*
- **Observe**
  - *Use Think Aloud Protocol*
- **Ask some questions**
  - *Debrief*
- **Write down what you learnt**

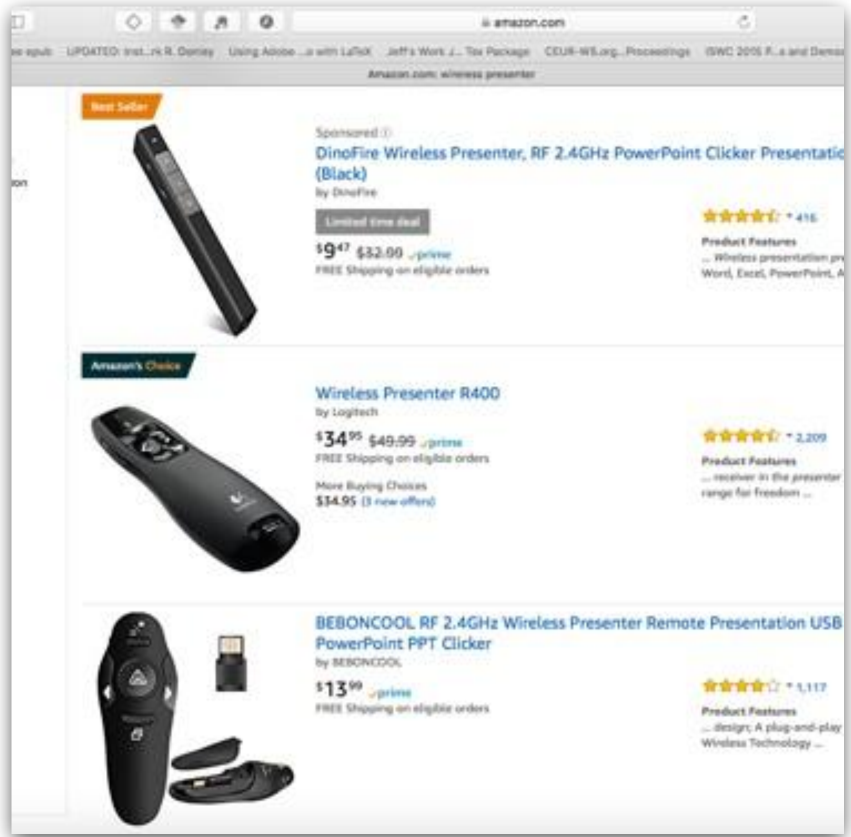
Potential Users

- **People who fall within the target audience**
  - *Attitudes*
  - *Behaviours*
  - *Characteristics*

- **Not current users**
  - *OK if current users of system but not for selected tasks*

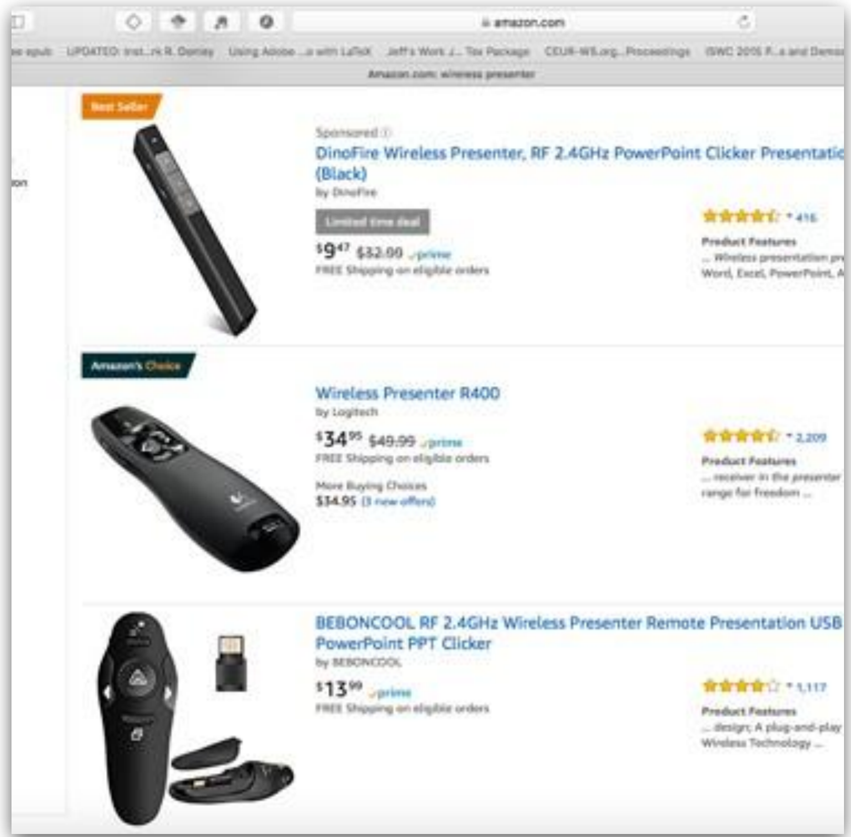
TASKS - What you ask them to do?

Buy a wireless presenter that costs less than $50 and has rechargeable battery.



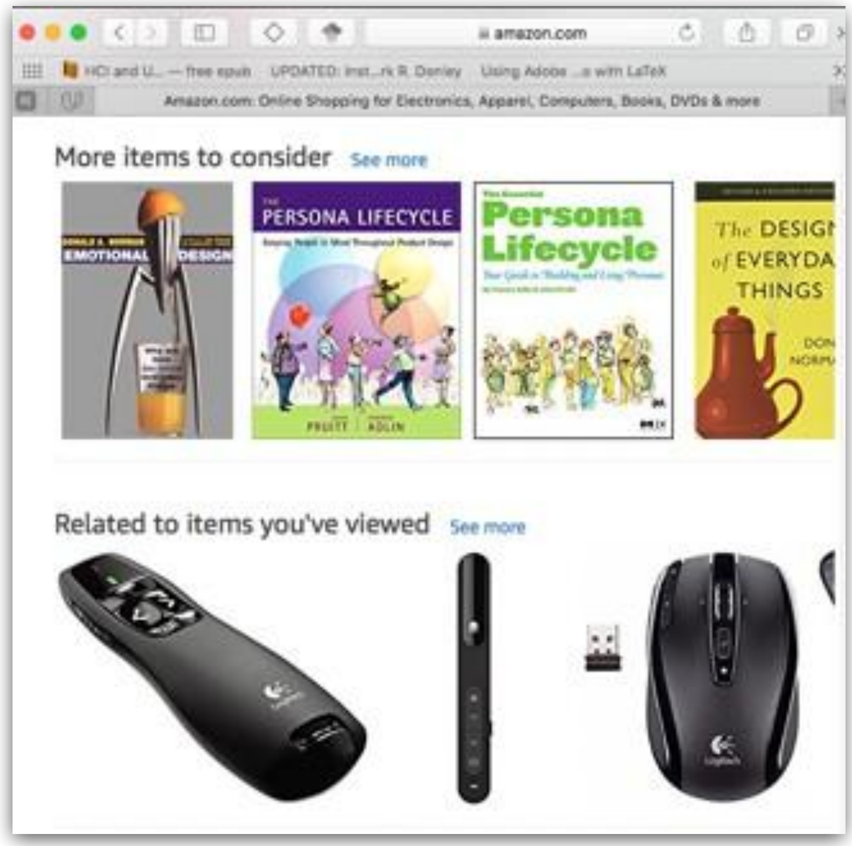
TASKS - What you ask them to do?

Enter a review for the most recent book you read



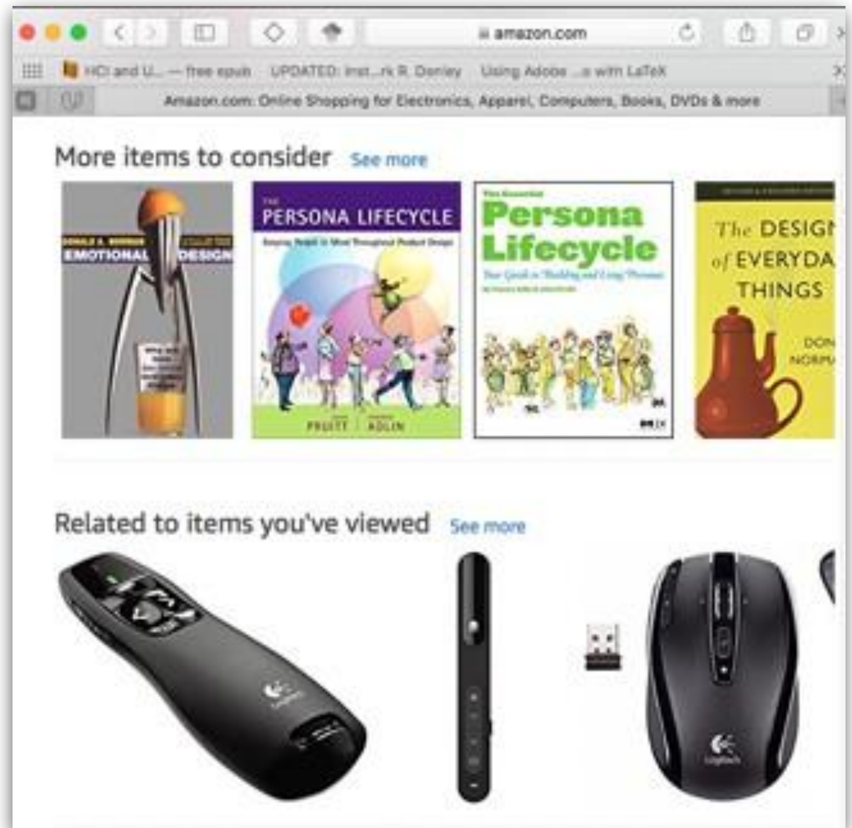
Choosing Tasks

- Things that most users need to do



Choosing Tasks

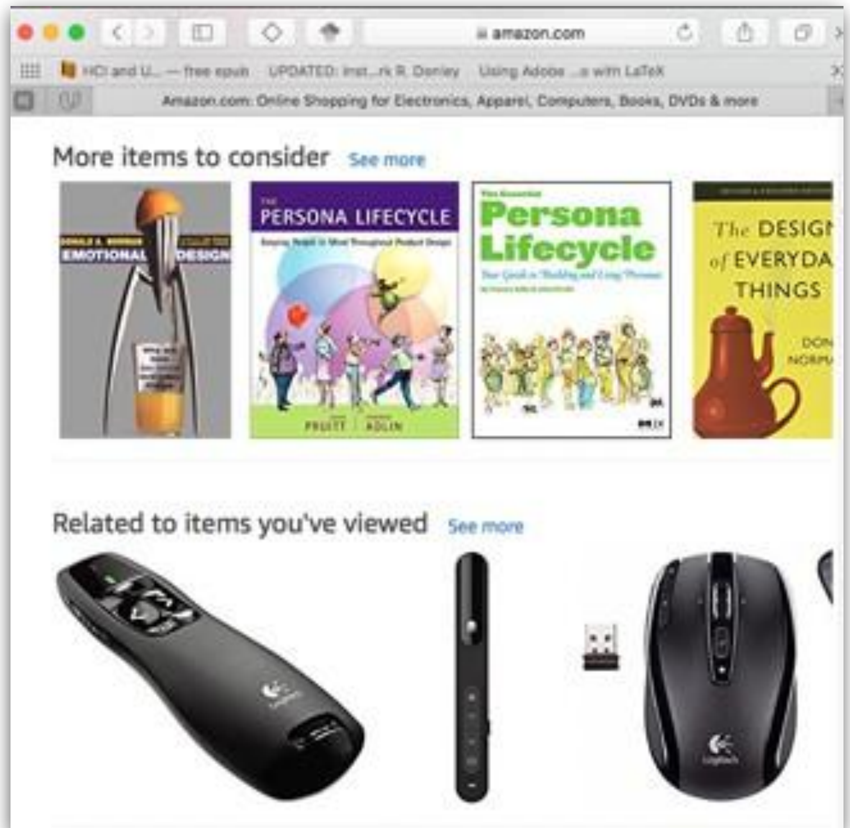
- Things that ***most*** users need to do
- More difficult things that ***some*** users need to do



Choosing Tasks

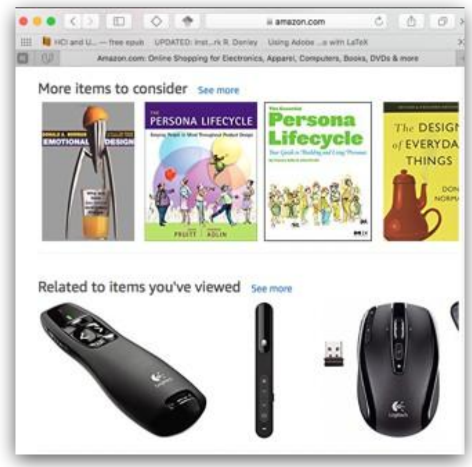
- Things that *most* users need to do
- More difficult things that *some* users need to do

Buy a hardbound book by a Nigerian author that was published this year.



Choosing Tasks: Task Types

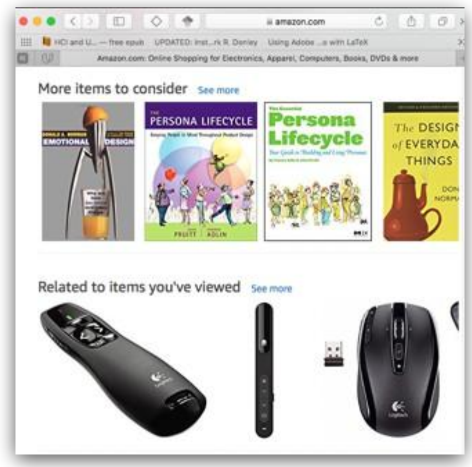
- **Closed-ended tasks**
  - *Have a clear end point*
  - *Have a verifiable outcome*
  - *Follow an expected path*
-



Buy a hardbound book by a Nigerian author that was published this year.

Choosing Tasks: Task Types

- Closed-ended tasks
  - *Have a clear end point*
  - *Have a verifiable outcome*
  - *Follow an expected path*
- **Open-ended tasks**
  - *Allow user to judge when complete*
  - *May not be verifiable*
  - *Allow following alternate paths*



Find some books you might like to read on vacation!

Which are better?

- **Closed Ended**
  - *Less Natural*
  - *Control for Motivation*
  - *Control for Interpretation*
  - *Assess success*
- **Open-Ended**
  - *More Natural*
  - *Varying motivation*
  - *Varying interpretation*
  - *Success?*

Task Sets

- **Progress from easier to harder**
  - *Why?*
- **Cover a range of critical task *types***
  - *e.g. browse, search, buy*
- **Can include open-ended and closed-ended tasks**
  - *Often a good idea!*
- **Be careful to avoid "*ordering effects*"**

Better Experimental Design for Better User Testing: http://www.uxbooth.com/articles/better-experimental-design-for-better-user-testing/

“User testing is a key part of validating design decisions, but what if the design of the test itself is invalid?

As a researcher, being aware of the pitfalls that affect experimental design ensures your test can lead to more elegant end user experiences.”

Task Design - Common Pitfalls

Is there a problem here:

Put 3 books in your shopping cart, then purchase them using Standard Shipping.

Task Wording

- **Don't lead the Witness**

Put 3 books in your shopping cart, then purchase them using Standard Shipping.

Choose three books and buy them, making sure they can get here by next Wednesday

Is there a problem here:

Use a list to find a gift for your ten year old nephew.

Task Wording

- **Avoid Ambiguity**

Use a list to find a gift for your ten year old nephew.

A friend told you that Amazon has helpful helpful gift idea list for different age groups.
Find a list that would be useful for finding a gift for your 10-year old nephew's birthday party, and select a gift for him.

Is there a problem here:

Buy a book and have it shipped by Monday!

Task Wording

- **Include context and motivation where needed**

Buy a book and have it shipped by Monday!

It's Friday, and you've just realized that you need a book on Japanese business etiquette by Monday, so that you have time to brush up before your meeting with colleagues from Japan on Wednesday. Find and purchase an appropriate book, making sure it arrives in time.

Task Wording

- **Include Information about how and when to stop**

It's Friday, and you've just realized that you need a book on Japanese business etiquette by Monday, so that you have time to brush up before your meeting with colleagues from Japan on Wednesday. Find and purchase an appropriate book, making sure it arrives in time.

When you feel you have completed the task, please say I'm done.

Task Wording

- **Pilot Test!**
  - *It's very hard to get right on the first try*
  - *Try the tasks out…*
    - Yourself
    - With a friend of two
    - …before trying them with real participants!

Think Aloud Protocol

- **Participants say (out loud) what they are thinking**
- **Thinking includes:**
  - *Looking for*
  - *something*
  - *Reading Text*
  - *Hypothesizing about how the system*
  - *might work Interpreting system options*
  - *Interpreting system*
  - *feedback Explaining*

    decisions

    Feeling frustrated, happy

Think Aloud Protocol

- **Not natural for a lot of people**
  - *Sometimes your participants will get quiet, they'll get absorbed in what they are doing*

- **Gently remind them to say hey, could you let me know what you're doing, what you're thinking?**
  - *Make sure you keep telling me what's going on, so that I can understand how you're going about this process.*

To Use or not to use Think Aloud?

- **Advantages:**
  - *Hear how the user thinks about the task*
  - *Learn what the user actually sees and notices*
  - *Hear how the user interprets options and feedback*

- **Disadvantages:**
  - *Timing will not be realistic*
  - *Attention to detail will not be quite realistic*
  - *Need to determine "rules of engagement" for questions, mistakes etc.*

Problem Finding: Using Think Aloud

Use Think
Aloud!

Debrief (After Tasks)

Debrief (After tasks)

- **Review problems, get more information**
- **Ask about usefulness, value**
  - *What would you find it useful for?*
  - *How would it fit in?*
  - *How would you value this?*
  - *Where would this provide a real benefit for you or*
  - *would it provide a benefit at all?*
- **Ask about perceived usability**
  - *Did they feel like it was usable?*
  - *The perception can sometimes be different from the observations that you made.*
- **You can ask them about the aesthetics, the credibility, and other aspects of UX**
- **Compare to the alternatives that they currently use**

Making sense of the Test!

- **Capturing "critical incidents" (things worth noting)**
  - *Errors*
    - People didn't do what you expected
    - Didn't follow the right path for accomplishing something
  - *Expressed frustration*
  - *Breakdowns*
  - *Pleasant surprises*
- **Assess overall success/failure**
  - *Usually a spectrum*

Making sense of the Test!

- **Capture overall reaction and reaction to specific aspects**
  - *What was the participant's demeanor?*
  - *Were they positive and upbeat and energetic while interacting with it or did they sort of seemed deflated or frustrated or bored?*
  - *And what were the causes of those different reactions if you can tease them out?*
- **Link incidents, success/failure and subjective reaction**
  - *Did they become frustrated and angry because of particular incidents that are connected to particular aspects of the design?*
  - *Or were there other reasons that that happened.*
  - *Were there specific incidents that led to a positive subjective reaction?*

Learning from the Test!

- **Quick! Write it down!**
- **Critical incidents, and why they happened**
  - *Mental model mismatches*
  - *Misinterpretation*
  - *Invalid assumptions made by the system*
  - *Missing user needs?*
  - *Too little flexibility*
  - *Too little guidance*

Learning from the test!

- **Problems => Severity: Impact on**
  - *Success/Failure*
  - *What was the impact on their subjective experience?*
  - *Did this problem lead to a change in their judgement of the overall experience of the system.*
  - *Product goals*
- **Other UX factors**
  - *Usefulness*
  - *Credibility*
  - *Accessibility*

One other very important thing!

- Participation is voluntary
- Participants can stop anytime
- Your are testing the system, not the participant
- You NEED to let the participants know this!

Whats "Micro" about this?

- **Relaxed recruiting**
  - *people that are close enough to target audience to be able to imagine*
  - *aka "hallway" usability test*
- **Fewer tasks (vs standard usability test)**
  - *< 30 min, rather than 60-90 min*
- **Little or no formal data collection**
  - *No recording*
  - *No questionnaires*
  - *No logging*
- **Off-the cuff analysis**